AI powered speech-to-speech mock interview platform

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Abstract

This project is an AI-powered speech-to-speech mock interview platform that helps engineering students prepare for job interviews through personalised voice-based practice sessions. The system takes a candidate's resume as input and uses a large language model to pull out key information like skills, projects, and work experience. This data is then used to create relevant interview questions on the fly, instead of using a fixed list of generic questions. Candidates reply verbally, and their answers are transcribed in real time using automatic speech recognition. The generated questions are turned back into natural-sounding speech, creating an entirely voice-driven, turn-based interview experience.

After each session, the platform assesses responses for clarity, relevance, and communication quality. It produces a structured feedback report that outlines strengths and areas for improvement. The platform uses a mix of local and cloud-based AI tools, including speech recognition, text-to-speech synthesis, and language models designed for specific tasks. It is lightweight, cost-effective, and can run on limited hardware. By merging resume-based personalisation with conversational voice interaction, this project seeks to improve the traditional methods of interview preparation, offering students a more realistic and engaging practice experience. This ultimately aims to enhance their readiness for competitive job markets.

Index terms

Mock interview system, Resume parsing, Speech-to-speech interaction

1. Introduction

Placement interviews are crucial for students transitioning from academics to professional jobs. However, many engineering students get limited chances for realistic, repeated interview practice before meeting recruiters. Traditional preparation methods, like peer-led mock interviews or faculty-guided sessions, face issues such as scheduling conflicts, evaluator availability, and inconsistent feedback. No two evaluators assess candidates in the same way. Meanwhile, the rise of large language models, speech recognition systems, and speech synthesis technologies makes it possible to create a realistic one-on-one interview experience without needing a human interviewer.

This project suggests an AI-powered, speech-to-speech mock interview platform that tailors the interview experience based on each candidate's resume. Instead of providing a standard set of questions, the system pulls structured information from the uploaded resume, including technical skills, academic projects, and previous work experience. It then generates interview questions that relate to the candidate's background. The candidate answers each question verbally, and the system transcribes, evaluates, and responds, creating a natural conversation rather than a text-based quiz.

The motivation for this project comes from a practical gap noticed in the institution's placement cell. Students often feel unprepared not due to a lack of technical skills but because they miss the specific rhythm and pressure of a spoken interview. A platform that can simulate this experience repeatedly, privately, and affordably meets a genuine and ongoing need.

2. Related Works

Interview preparation tools have changed a lot over the past decade. They have shifted from simple question banks to more adaptive, AI-driven systems. Early digital tools mainly provided lists of common interview questions based on role or company. They offered no personalisation or feedback. These tools helped candidates learn about question types but did not replicate the pressure of a real conversation.

Another area of focus has been automated resume screening and parsing, mainly created for applicant tracking systems used by recruiters. These systems use techniques like named entity recognition and structured information extraction to turn unstructured resume text into organised fields, such as skills, education, and experience. While they are good for filtering candidates on a large scale, these systems generally serve employers, extracting information without helping candidates prepare.

What remains less explored is the mix of resume-based personalisation with a fully voice-driven, turn-based conversation interface. This approach should be affordable and work on limited student hardware rather than enterprise systems. Most existing tools either personalise without voice interaction or allow voice interaction without connecting the conversation to the candidate's real background. This project aims to fill that gap by combining resume-aware question generation with a natural speech interface, all at a cost accessible to students and the institution's placement office.

3. Proposed System

The proposed system is a modular pipeline that turns a candidate's resume and spoken responses into a complete, evaluated mock interview session. The design is intentionally divided into independent functional modules. This allows for the development, testing, and improvement of each module without affecting the others. It also enables upgrades to individual components, such as moving from local to cloud-based speech models, without redesigning the entire system.

3.1 System Architecture Overview

At a high level, the system follows a sequential pipeline: a candidate uploads a resume, the system extracts and organises relevant information from it, a set of personalised interview questions is generated, and the candidate goes through a turn-based spoken interview. This consists of the system asking a question out loud, the candidate responding out loud, and the system transcribing and evaluating that answer before moving to the next question. After completing the session, all responses are compiled into a structured feedback report. A backend orchestration layer manages the handoff between each module, maintaining session state throughout the interview.

3.2 Resume Parsing Module

The resume parsing module accepts a candidate's resume in PDF or DOCX format and extracts raw text, which is then sent to a language model that returns structured output identifying key fields, such as technical skills, project titles and descriptions, work experience, and educational background. This structured representation provides the basis for personalising every subsequent stage of the interview. It ensures that generated questions reference the candidate's actual projects and skills, rather than generic role-based templates.

3.3 Question Generation Module

Using the structured resume data as a guide, this module creates a sequence of interview questions tailored to the candidate's specific background. Questions cover a mix of resume-specific prompts, such as asking the candidate to elaborate on a listed project, and general role-appropriate questions relevant to the candidate's target field. Prompt engineering techniques help guide the language model toward realistic and professionally-worded questions instead of overly literal restatements of resume content.

3.4 Speech Processing Module

This module manages both directions of spoken communication. Generated questions are converted into natural-sounding speech using a text-to-speech engine and played back to the candidate. The candidate's spoken response is captured through the browser's audio recording interface and converted into text using an automatic speech recognition engine. This module operates in a turn-based manner, where each exchange is completed before the next begins, prioritising reliability over real-time conversational overlap.

3.5 Answer Evaluation and Feedback Module

Once a response has been transcribed, it is assessed using a defined rubric (A rubric is a scoring guide that outlines
specific criteria for evaluating a project, paper, or test.) that covers clarity, relevance to the question, and communication quality, including factors such as filler word frequency. A language model, prompted with the rubric criteria, performs this evaluation, producing a score and brief qualitative feedback for each answer. At the end of the session, all individual evaluations are combined into a single summary report presented to the candidate.

4. Expected Outcome

The conclusion of this project will result in the team presenting a working prototype that can perform a full resume-based voice interview from beginning to end. This includes the ability to transcribe, design questions and provide meaningful feedback. The system will have to be able to compete against generic interview preparation methods used today and will also have to be usable on standard student hardware at minimal cost. Besides the practical deliveries, the project is expected to benefit students preparing for placements at the organisation's placement cell.

5. References

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